Screenshots

➢ Membership functions

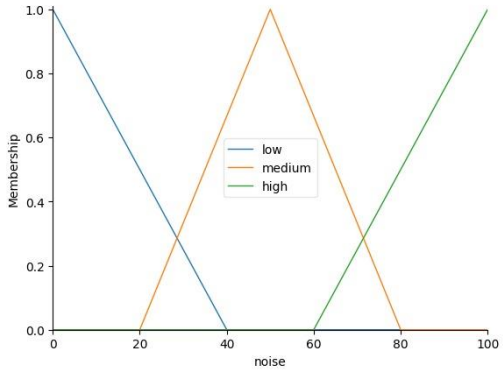
1. Noise membership graph

Step 4: Define Membership Functions

Ambient Noise Level

```
noise['low'] = fuzz.trimf(noise.universe, [0, 0, 40])
noise['medium'] = fuzz.trimf(noise.universe, [20, 50, 80])
noise['high'] = fuzz.trimf(noise.universe, [60, 100, 100])
```

```
noise.view()
```

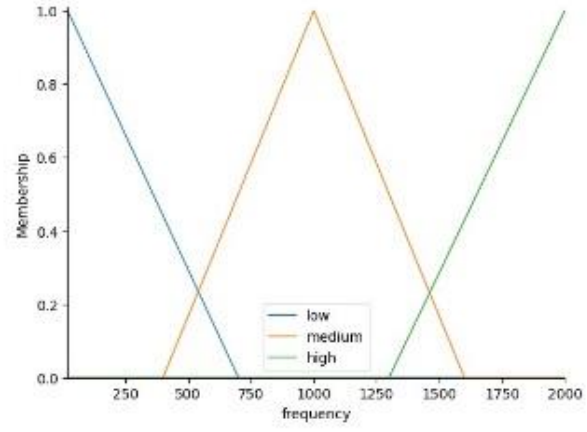


2. Frequency membership graph

Collapse 2 child cells under Noise Frequency (Press <Shift> to also collapse sibling sections)

```
frequency['low'] = fuzz.trimf(frequency.universe, [20, 20, 700])
frequency['medium'] = fuzz.trimf(frequency.universe, [400, 1000, 1600])
frequency['high'] = fuzz.trimf(frequency.universe, [1300, 2000, 2000])
```

```
frequency.view()
```

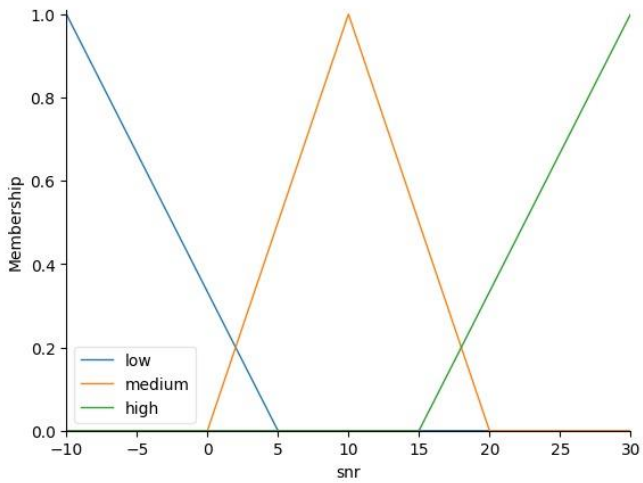


3. SNR membership graph

Signal-to-Noise Ratio

```
snr['low'] = fuzz.trimf(snr.universe, [-10, -10, 5])
snr['medium'] = fuzz.trimf(snr.universe, [0, 10, 20])
snr['high'] = fuzz.trimf(snr.universe, [15, 30, 30])
```

```
snr.view()
```

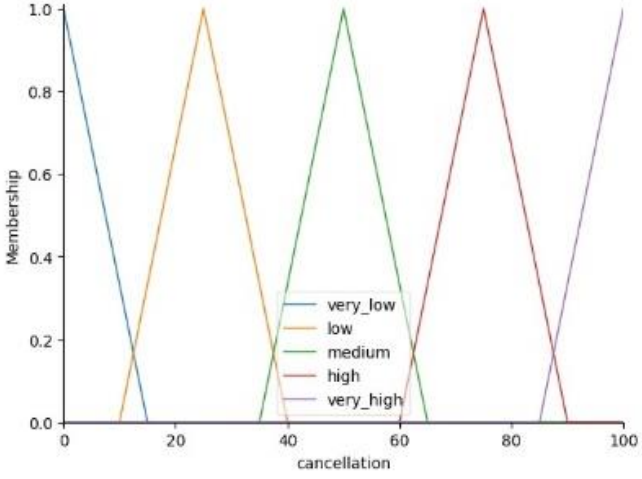


4. Cancellation membership graph

Noise Cancellation Gain

```
cancellation['very_low'] = fuzz.trimf(cancellation.universe, [0, 0, 15])
cancellation['low'] = fuzz.trimf(cancellation.universe, [10, 25, 40])
cancellation['medium'] = fuzz.trimf(cancellation.universe, [35, 50, 65])
cancellation['high'] = fuzz.trimf(cancellation.universe, [60, 75, 90])
cancellation['very_high'] = fuzz.trimf(cancellation.universe, [85, 100, 100])
```

```
cancellation.view()
```



➢ 10 test cases and their outputs.

Phase 4 - Testing

The fuzzy controller is tested using 10 different combinations of ambient noise level, noise frequency and signal-to-noise ratio (SNR).

The purpose of testing is to observe how the recommended noise cancellation gain changes for different environmental conditions.

```
# Use centroid method for testing
cancellation.defuzzify_method = 'centroid'

test_cases = [
    (20, 300, 15),
    (30, 500, 5),
    (40, 300, 15),
    (50, 1000, 10),
    (50, 1500, 5),
    (60, 1500, 5),
    (70, 900, 5),
    (70, 1500, 20),
    (80, 1500, 5),
    (90, 1000, 10)
]

print("Test Case | Noise(dB) | Frequency(Hz) | SNR(dB) | Cancellation Gain(%)")
print("-" * 78)

results = []

for i, (n, f, s) in enumerate(test_cases, 1):
    test_system = ctrl.ControlSystemSimulation(cancellation_ctrl)

    test_system.input['noise'] = n
    test_system.input['frequency'] = f
    test_system.input['snr'] = s

    test_system.compute()

    result = test_system.output['cancellation']
    results.append((i, n, f, s, result))

    print(f"{i:9} | {n:9} | {f:13} | {s:7} | {result:20.2f}")
```

```
Test Case | Noise(dB) | Frequency(Hz) | SNR(dB) | Cancellation Gain(%)
------------------------------------------------------------------------------
        1 |        20 |           300 |      15 |                 5.83
        2 |        30 |           500 |       5 |                20.06
        3 |        40 |           300 |      15 |                25.00
        4 |        50 |          1000 |      10 |                25.00
        5 |        50 |          1500 |       5 |                34.82
        6 |        60 |          1500 |       5 |                34.82
        7 |        70 |           900 |       5 |                47.03
        8 |        70 |          1500 |      20 |                39.90
        9 |        80 |          1500 |       5 |                75.00
       10 |        90 |          1000 |      10 |                75.00
```

➢ Defuzzification comparison

1. Centroid

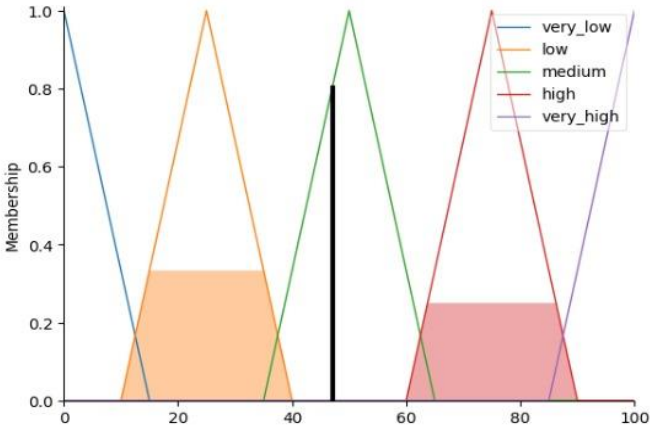
It finds the balance point of the aggregated fuzzy output.

```
cancellation.defuzzify_method = 'centroid'
system = ctrl.ControlSystemSimulation(cancellation_ctrl)
system.input['noise'] = 70
system.input['frequency'] = 900
system.input['snr'] = 5
system.compute()
print(round(system.output['cancellation'],2),"%")
```

```
47.03 %
```

```
cancellation.view(sim=system)

plt.show()
```



Using For Loop

```
methods = ['centroid', 'bisector', 'mom', 'som', 'lom']

for method in methods:
    cancellation.defuzzify_method = method

    system = ctrl.ControlSystemSimulation(cancellation_ctrl)

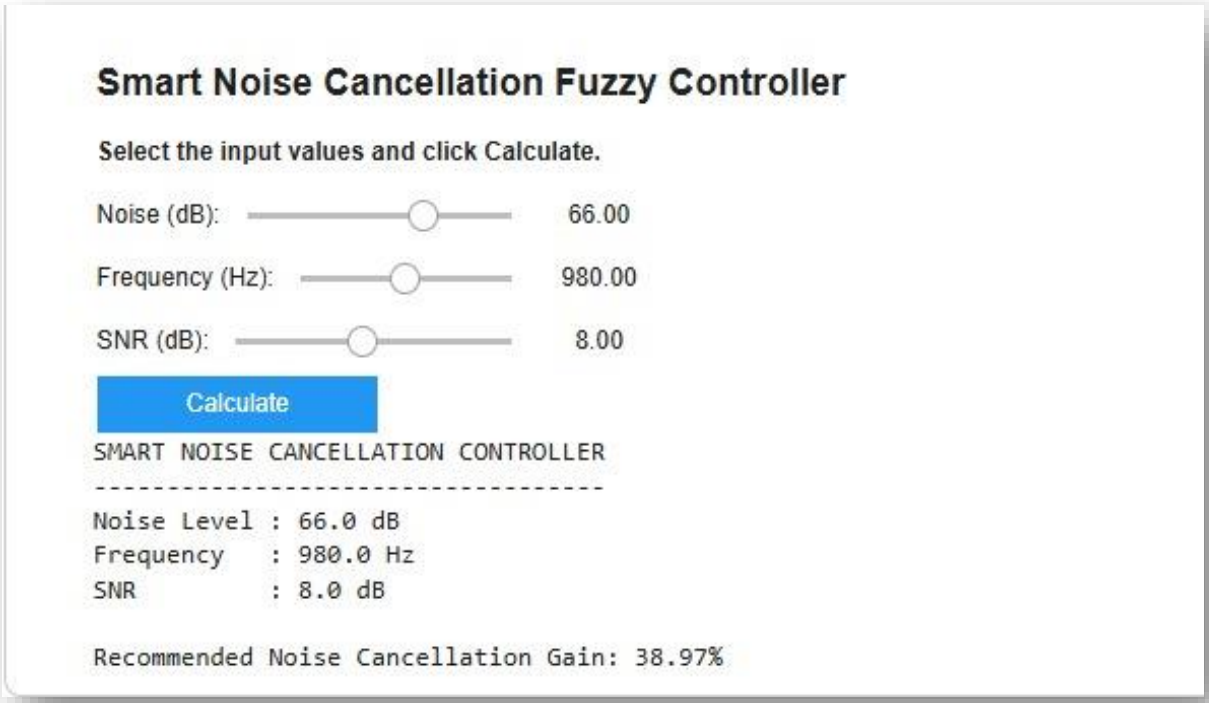
    system.input['noise'] = 70
    system.input['frequency'] = 900
    system.input['snr'] = 5

    system.compute()

    print(method, system.output['cancellation'])
```

```
centroid 47.02797202797205
bisector 34.84374999999999
mom 25.0
som 15.0
lom 35.0
```

➢ UI


Smart Noise Cancellation Fuzzy Controller
Select the input values and click Calculate.
Noise (dB):
66.00
Frequency (Hz):
980.00
SNR (dB):
8.00
Calculate
SMART NOISE CANCELLATION CONTROLLER
Noise Level : 66.0 dB
Frequency : 980.0 Hz
SNR : 8.0 dB
Recommended Noise Cancellation Gain: 38.97%